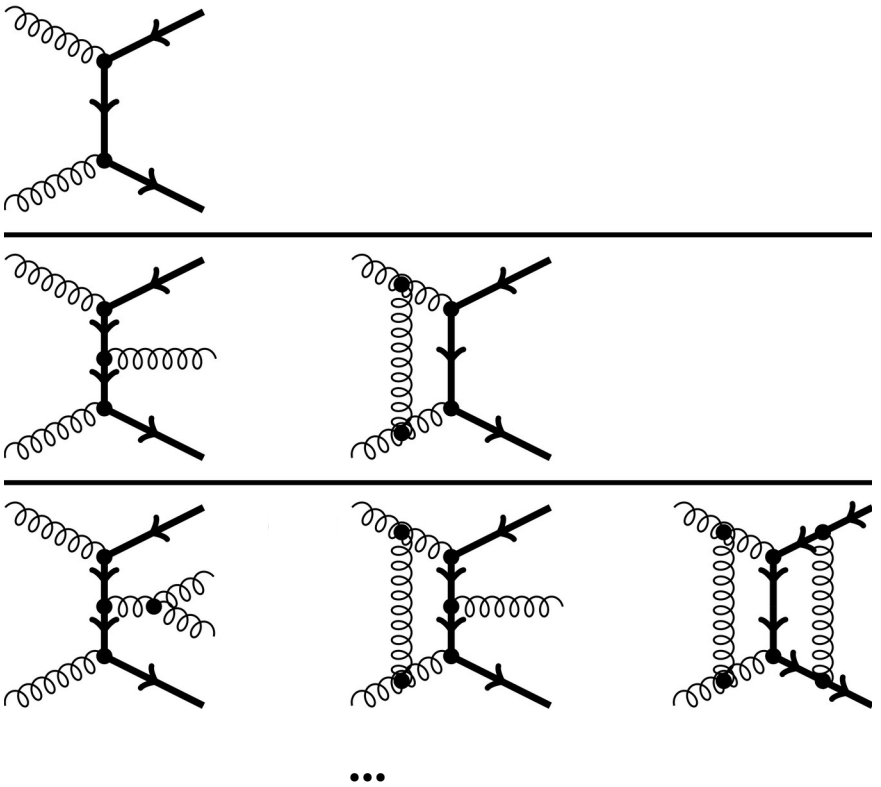
Quantum Chromodynamics

René Poncelet

Lecture 1:
the QCD Lagrangian, the strong coupling constant,
Final state radiation & IR divergencies

THE HENRYK NIEWODNICZAŃSKI
INSTITUTE OF NUCLEAR PHYSICS
POLISH ACADEMY OF SCIENCES

European School of High Energy Physics
19th June - 3rd Jul 2026 - Ostroda



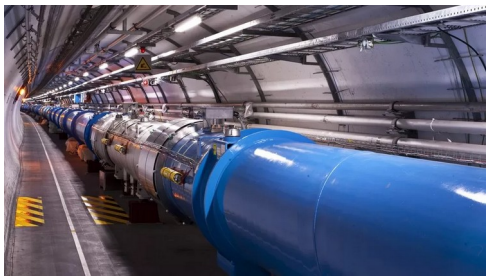
Lecture outline

- **Motivation**
- **The QCD Lagrangian**
  + running of the strong coupling
- **Infrared divergencies**
- Parton-distribution functions
- Anatomy of a higher-order QCD computation
  - Virtual/Loop corrections
  - Real emission corrections
- Phenomenology

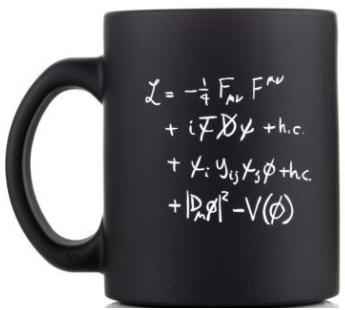
What are the fundamental building blocks of matter?

Scattering experiments

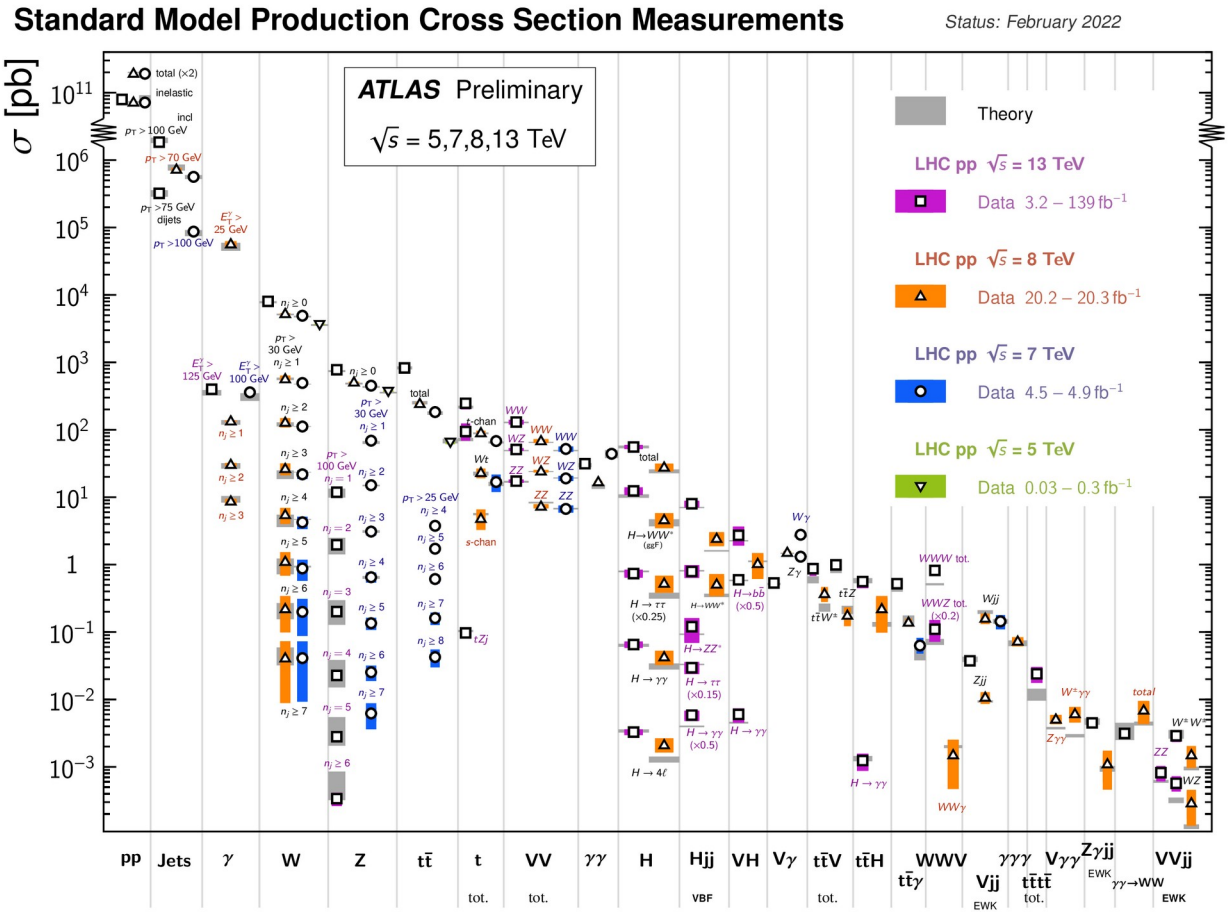
Large Hadron Collider (LHC)



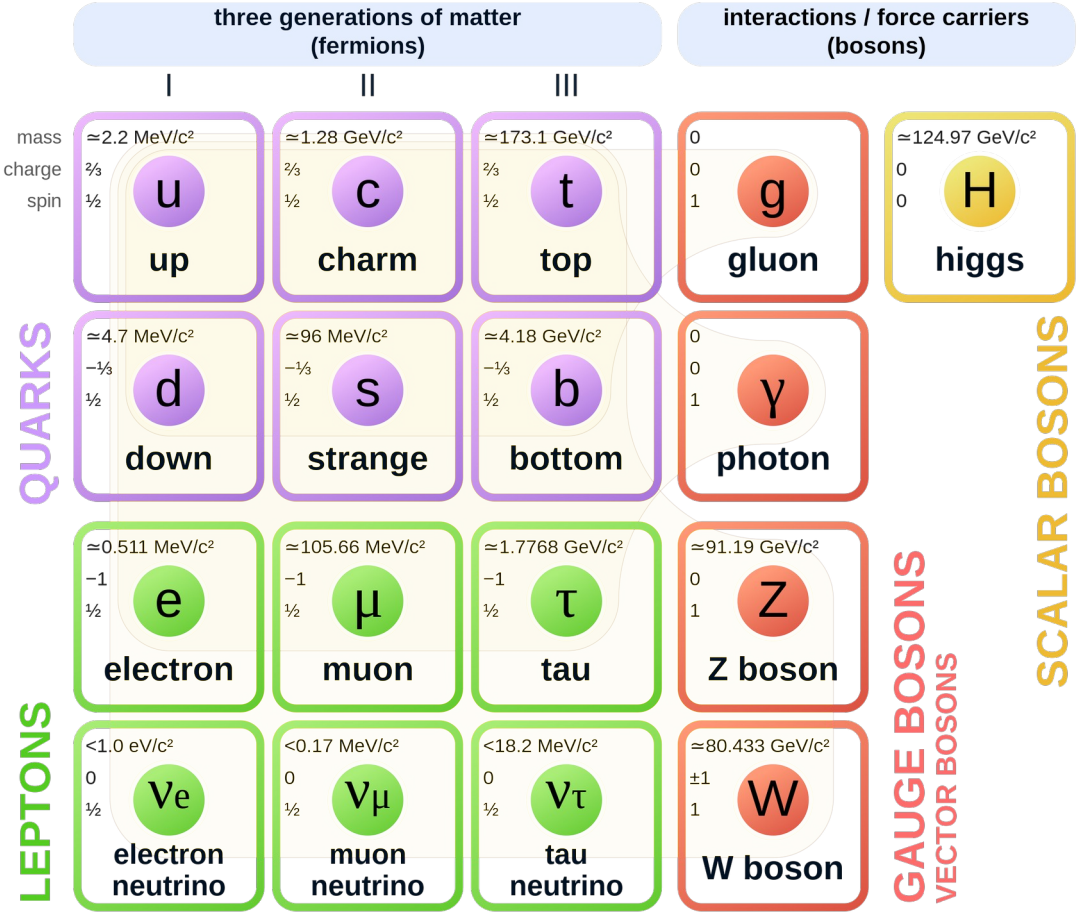
Credit: CERN

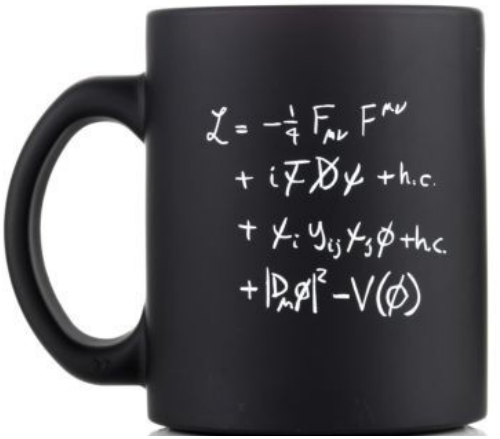


Theory/ Standard Model



Standard Model of Elementary Particles

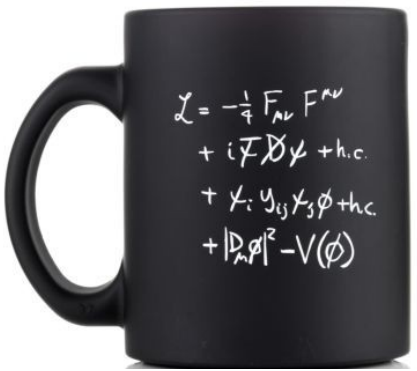


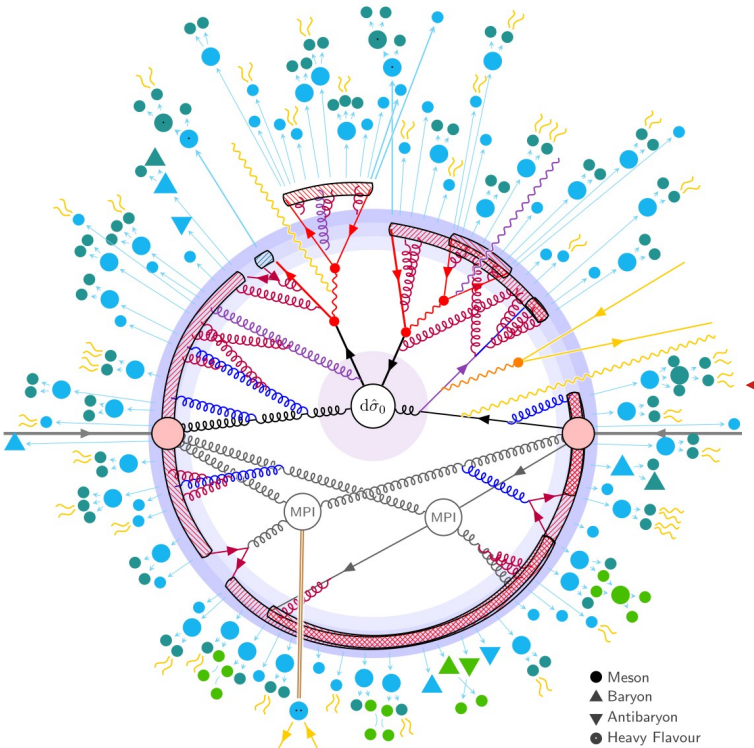


Credit: Wikipedia/CERN

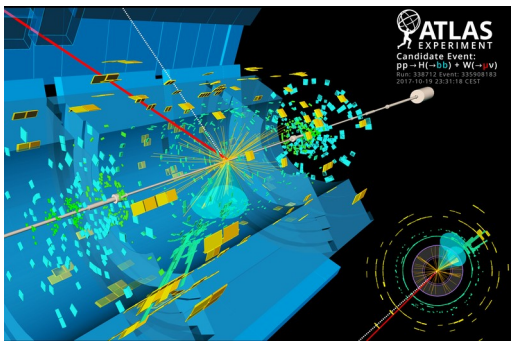
Monte Carlo Event Generators

… the **bridge** between “theory world” and “real world”





[Pythia 8.3]



Scale factorization in collision events

Guiding physics principle
"What you see depends on the resolution"

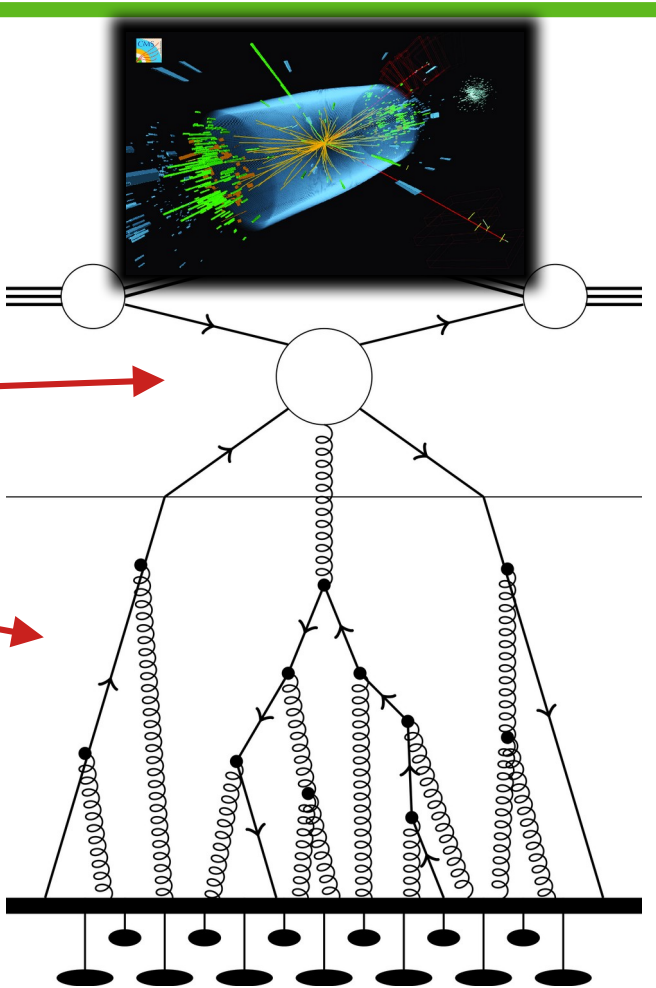
Example: Quantum Chromodynamics (QCD)

$Q \gg \Lambda_{\mathrm{QCD}}$ **Fixed-order perturbation theory**

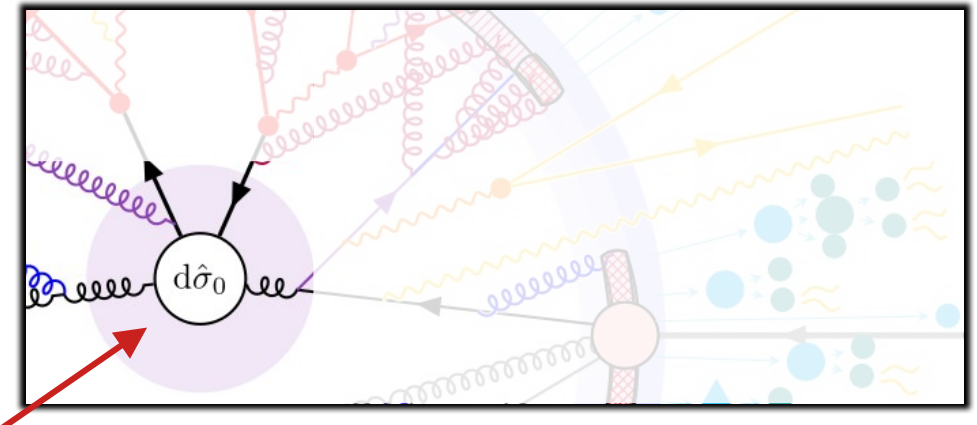
$Q \gtrsim \Lambda_{\mathrm{QCD}}$ **Parton-shower/Resummation**

$Q \sim \Lambda_{\mathrm{QCD}}$ **"Hadronization"/MPI/…**

Energy



The tasks of QCD simulation at colliders

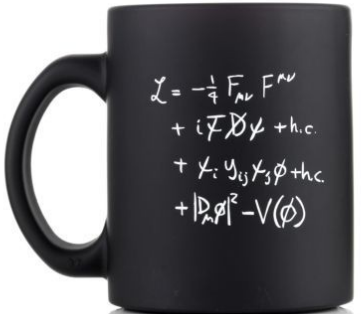


Energy

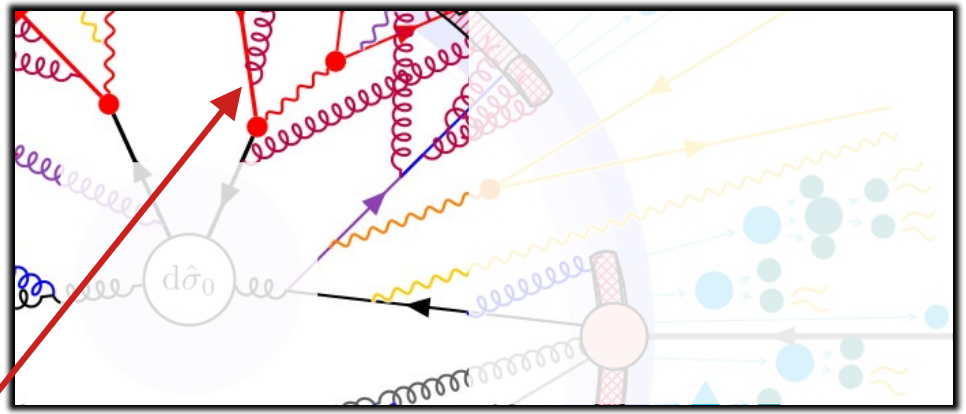
$Q \gg \Lambda_{\mathrm{QCD}}$

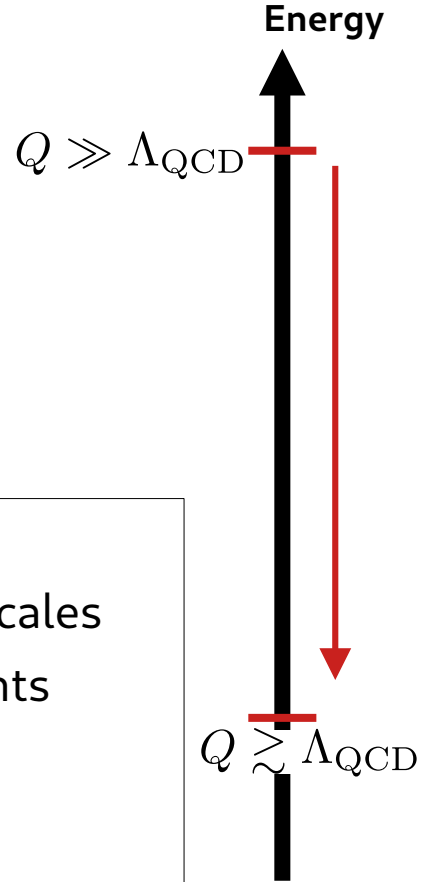
Hard scattering process

- Individual partons scatter at high momentum transfer
- Matrix elements directly from Lagrangian
- Fixed-order perturbation theory controls precision
  → expansions in orders of QCD and EW couplings



The tasks of QCD simulation at colliders

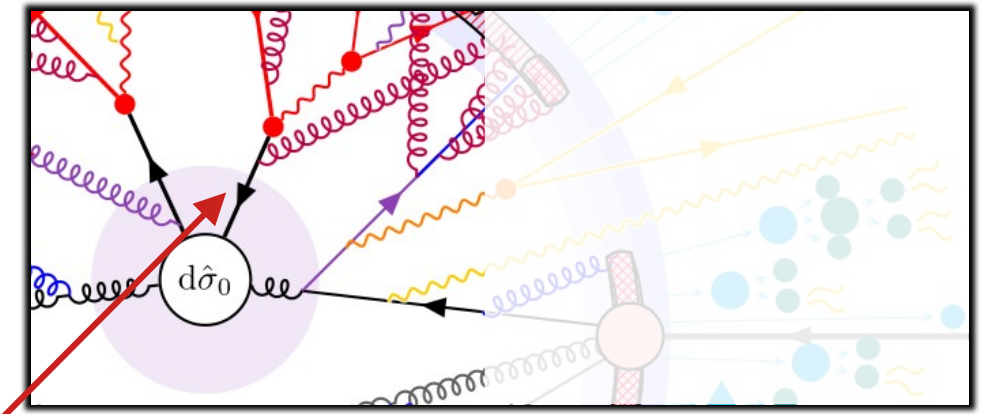


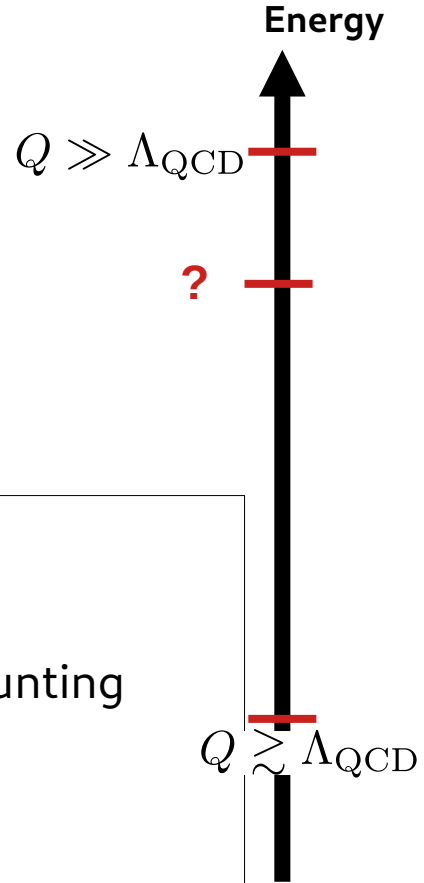


Parton-shower evolution

- Particle radiation: sequence of parton splittings at successively lower scales
- Splitting probabilities → soft/collinear approximations of matrix elements
- all-order resummation, accuracy measured in resummed logarithms
- Accuracy controlled by:
  splitting functions, kinematics, treatment of color & spin

The tasks of QCD simulation at colliders

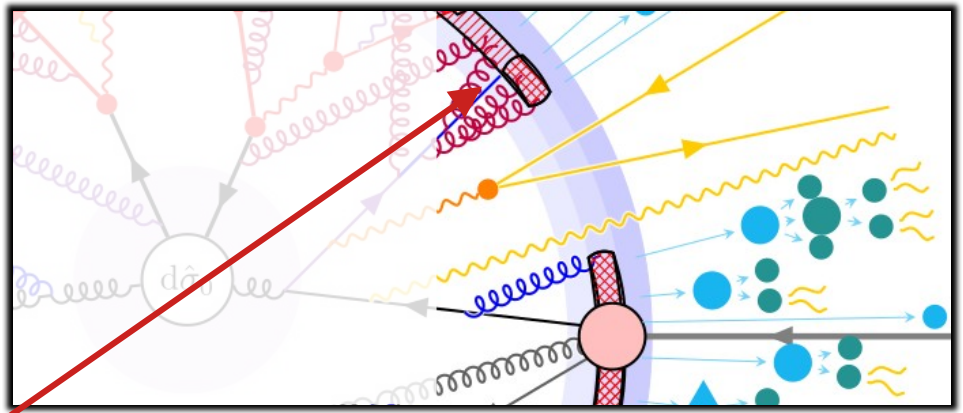




Matching/Merging

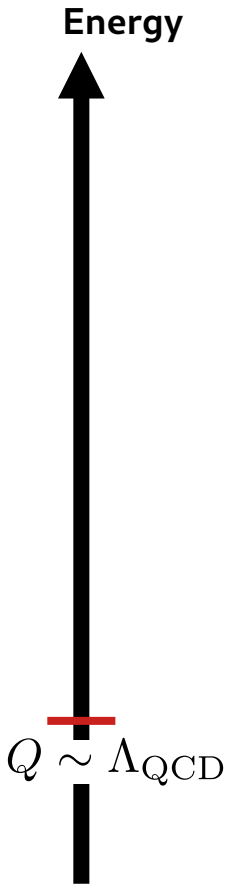
- combine hard scattering and parton-shower
- tricky bit: how to preserve the **precision/accuracy** and avoid double counting
- solve pathologies of higher fixed-order computations

The tasks of QCD simulation at colliders



Hadronization

- Formation and decay of observable particles
- Intrinsically non perturbative physics
  → driven by phenomenological models
  → input from experiments



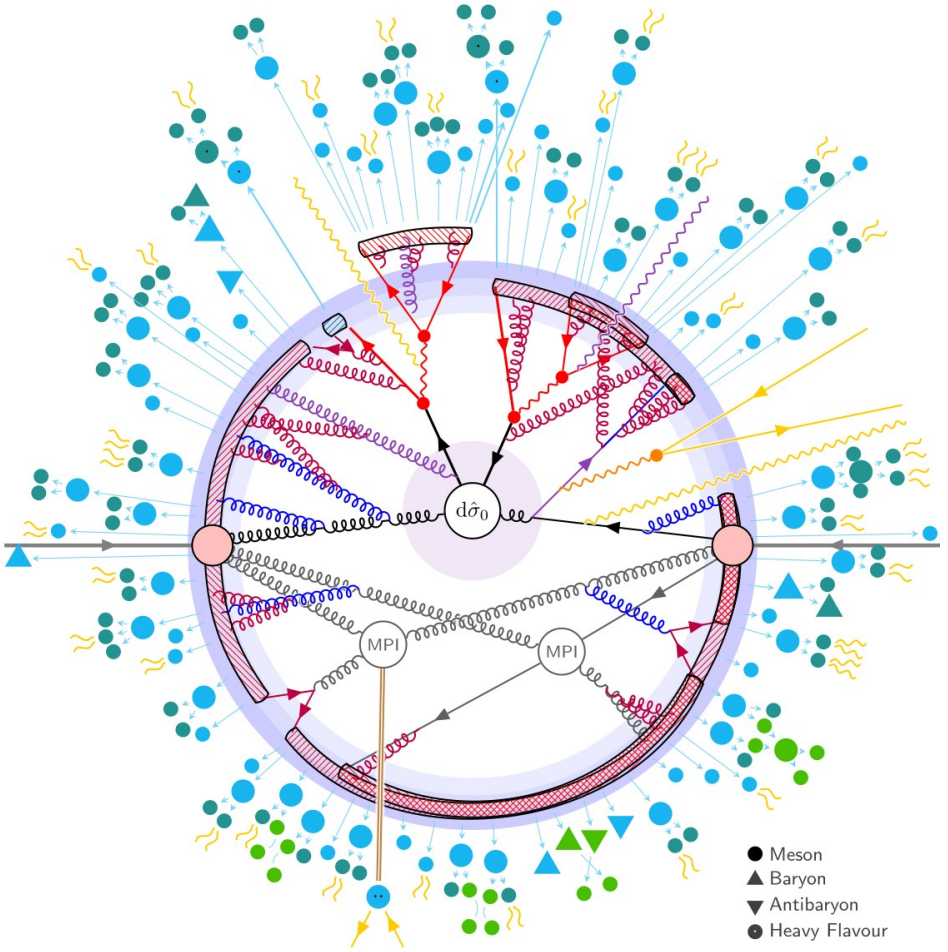
The tasks of QCD simulation at colliders

Global event considerations

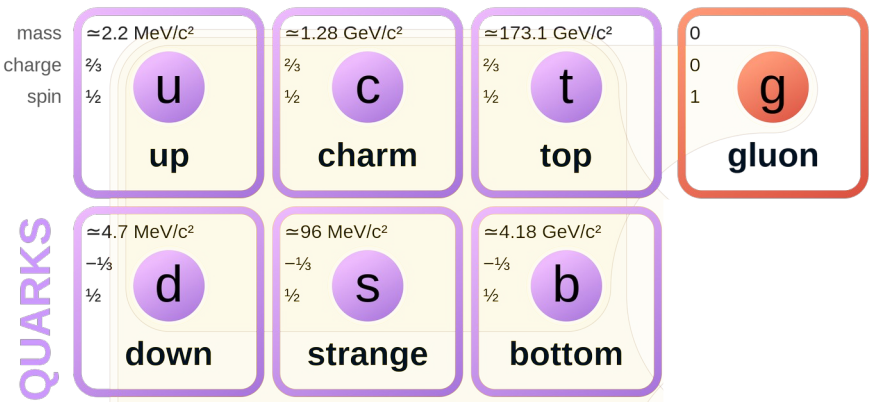
- underlying event
- color reconnection
- multi parton interactions (MPI)
- beam remnants
- …

Technical aspects

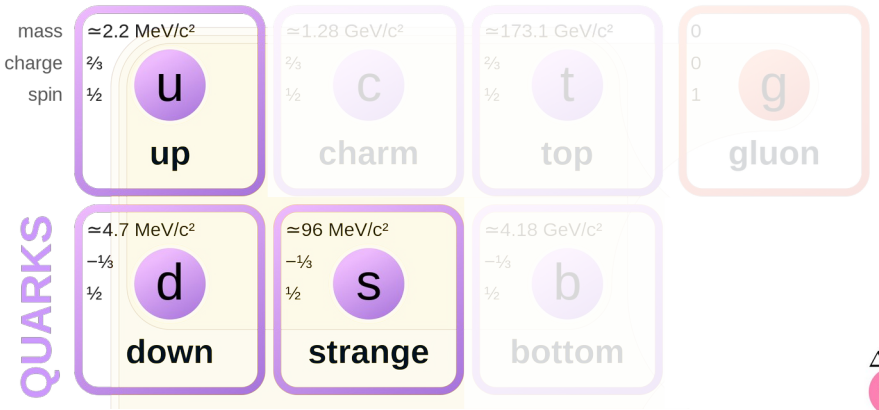
- Performance
- Tuning
- Uncertainties



A bit of history...

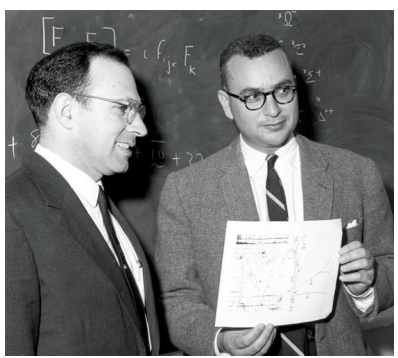


The eight-fold way

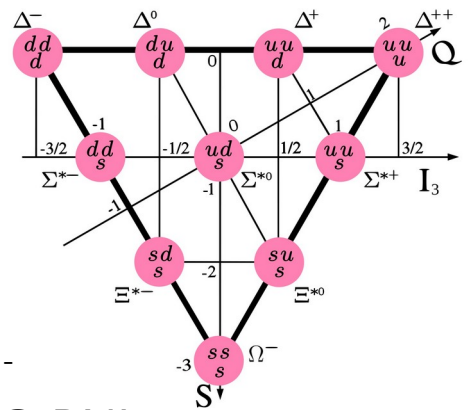


Early 60's: a zoo of heavy spin 3/2 particles
→ Gell-Mann/Zweig's proposal:
build up from 3 spin ½ particles
→ called Quarks (u,d,s)

[Finnegans Wake von James Joyce („Three quarks for Muster Mark")]



Yuval Ne'eman (left) and Gell-Mann
March 1964 (CERN Courier'19)



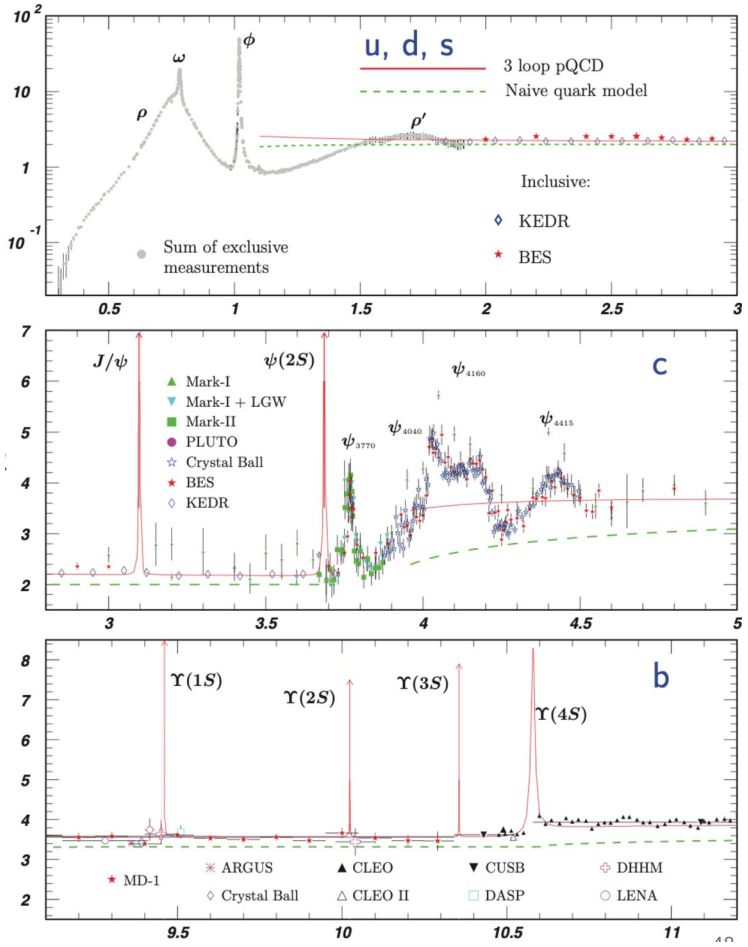
Prediction of Ω^-
→ found in '64 @ BNL

+ existence of light quarks confirmed
in '68 by DIS experiments at SLAC

However:
qqq states seem to violate
Pauli's exclusion principle!

Postulate
new **colour** quantum number!

Confirmation on three colours



Experimental evidence for colour:

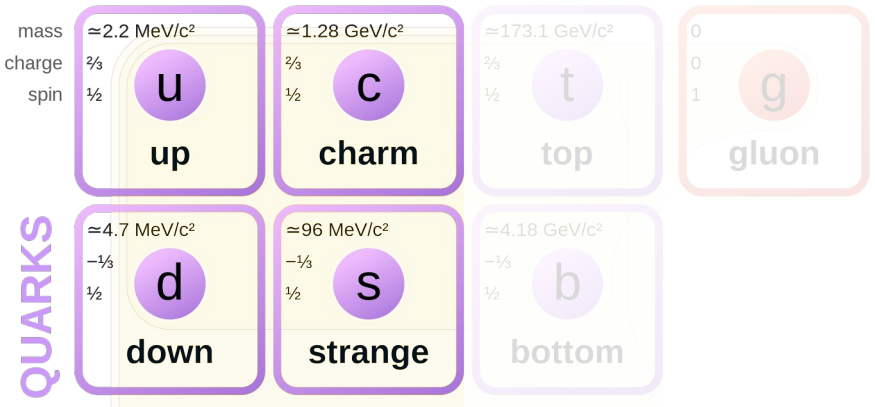
$$R = \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)} = N_c \sum_q Q_q^2 \theta(\sqrt{s} - 2m_q)$$

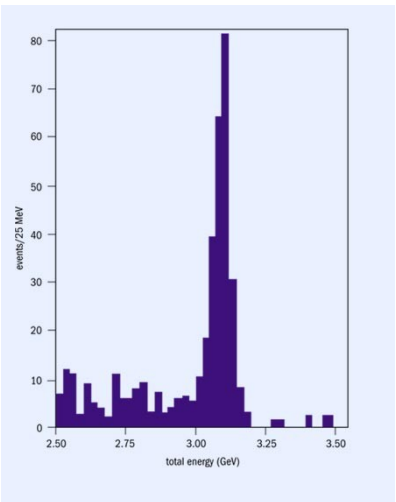
Comparison to data (away from resonances):

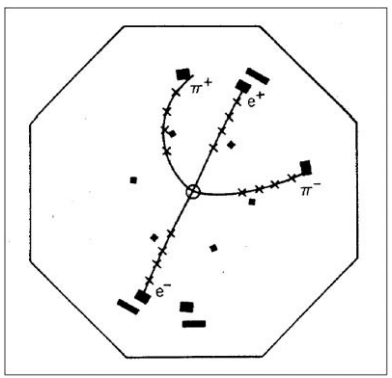
$$\rightarrow \quad N_c = 3$$

But also indirect evidence through jet measurements
→ sensitive to the QCD emission pattern

The charming 4th quark



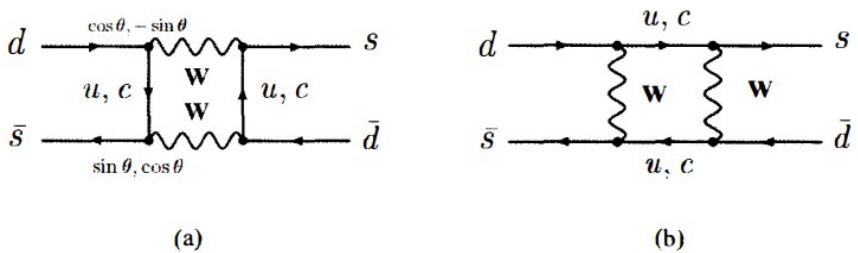




[CERN Courier'04]

Glashow, Ilioppoulos, Maiani'70

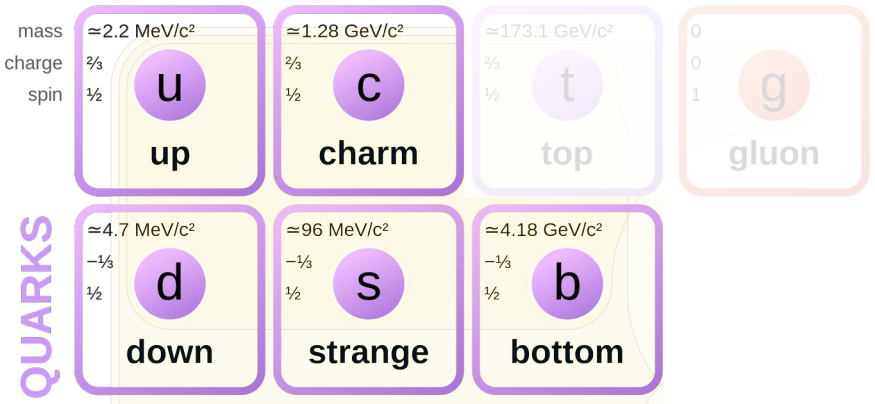
GIM mechanism
suppression of flavour neutral currents through interference with 4th quark!



$$\sim m_u^2 - m_c^2$$

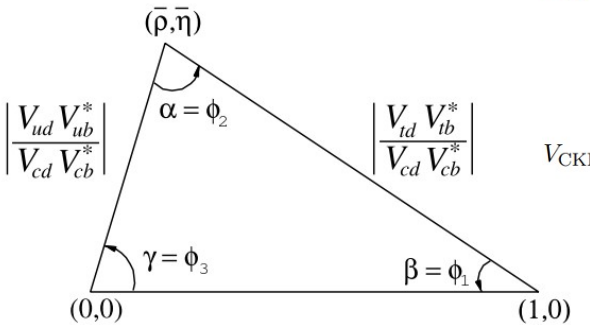
Charm quark directly produced at SLAC and BNL as J/Psi bound state '74

CP violation and the bottom quark

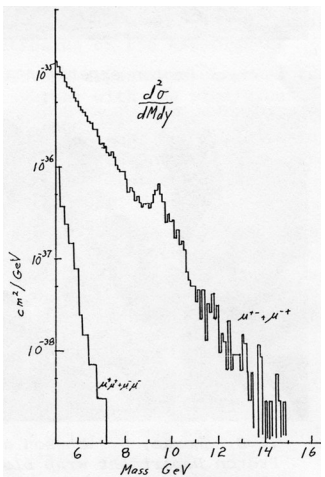


Kobayashi & Maskawa ‘73:
description of CP violation (via CKM matrix)
→ required a third generation of quarks

$$V_{\mathrm{CKM}} \equiv V_L^u V_L^{d\dagger} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix}$$



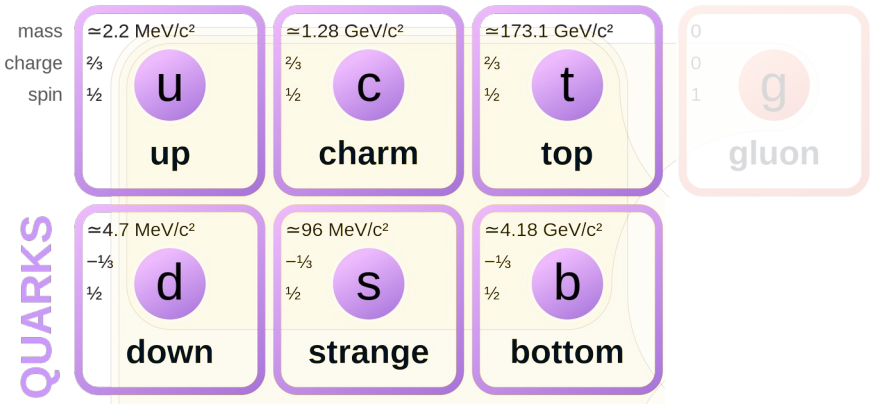
$$V_{\mathrm{CKM}} = \begin{pmatrix} 1-\lambda^2/2 & \lambda & A\lambda^3(\rho-i\eta) \\ -\lambda & 1-\lambda^2/2 & A\lambda^2 \\ A\lambda^3(1-\rho-i\eta) & -A\lambda^2 & 1 \end{pmatrix} + \mathcal{O}(\lambda^4)$$



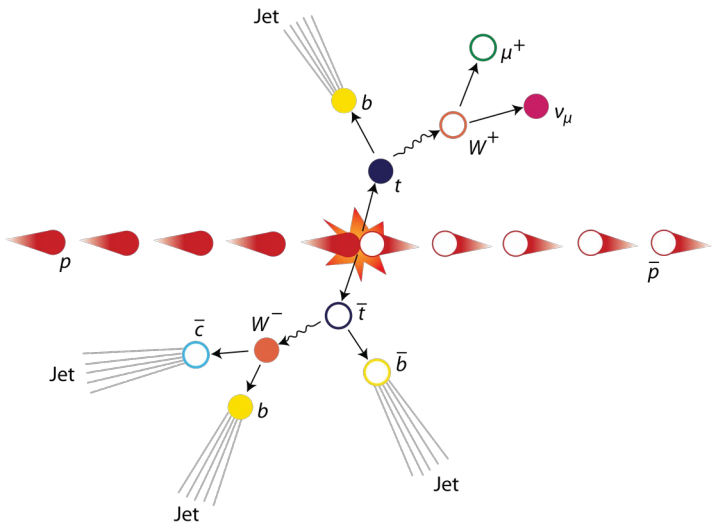
[E228 experiement’77]

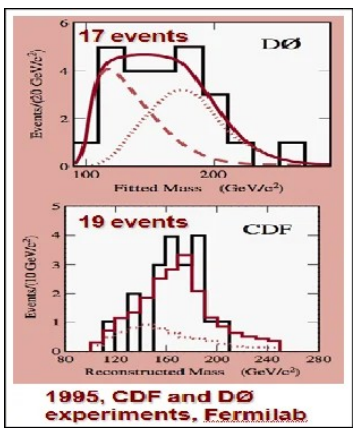
→ direct discovery of the Upsilon @ FermiLab ‘77

Reaching the top



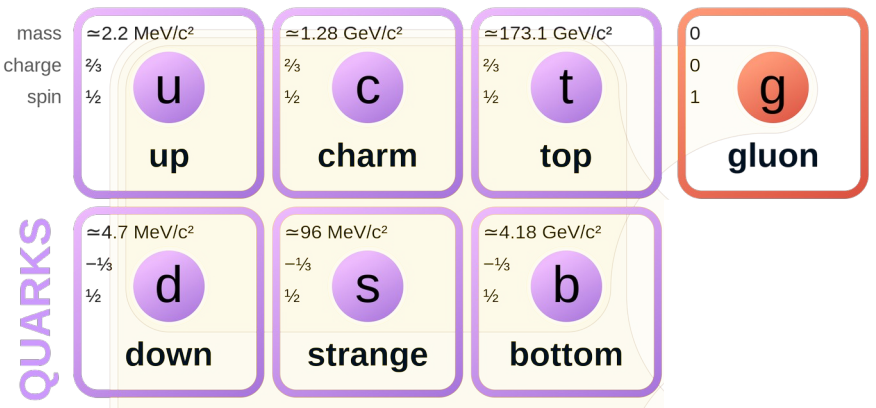
Indirect evidence:
- CKM matrix
- B meson oscillations [ARGUS'87]
- Precision EW measurements
→ top-quark has to be heavy ~170 GeV





Direct production at the Tevatron
by CDF & D0 at FermiLab

The QCD Lagrangian



Building up the QCD Lagrangian

1. ingredient: fermions (aka quarks)

Free quark Lagrangian: $\mathcal{L}_q^0 = \sum_q \bar{\Psi}_{q,a}(i\partial\!\!\!/\delta_{ab} - m_q\delta_{ab})\Psi_{q,b}$

one copy for each colour

$q = d, u, s, c, b, t$

quark masses are free parameters in QCD
→ in the SM they are determined by the electro-weak sector

a free theory is quite boring → add some interactions!

But how?
→ learn from QED as a local gauge theory
→ interpret colour as charge under local gauge group: SU(3)

Building up the QCD Lagrangian

2. ingredient: introduce SU(3) symmetry

→ impose symmetry transformation that relates N_c different colours → SU(N_c)

$$\Psi_a \to \Psi'_i = U_{ij}(x)\Psi^j \ , \quad \bar{\Psi}_a \to \bar{\Psi}'_i = \bar{\Psi}^j U^{-1}_{ji}(x)$$

Unitary → N_c^2-1 = 8 free parameters

Must be smoothly connected to unity → express in terms of **generators** close to it!

$$U_{ij}(x) = \exp\left\{ i \sum_{A=1}^{N_c^2-1} t^A \theta^A(x) \right\}_{ij} = \delta_{ij} + i \sum_{A=1}^{N_c^2-1} t^A_{ij} \theta(x)^A + \mathcal{O}(\theta^2)$$

Gell-Mann matrices

The Gell-Mann matrices are a matrix representation for $N_c = 3$: $t^a_{ij} = \lambda^a_{ij}/2$

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad \lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \quad \lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad \lambda^8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

Local gauge symmetry

3. ingredient: covariant derivate

How do quark fields transform?

→ mass term, no problem!

→ derivative? $\not{\partial}(U(x)\Psi(x)) = U(x)\not{\partial}\Psi(x) + (\not{\partial}U(x))\Psi(x)$

Solution (same idea as in QED): introduce **covariant derivative**

$$D_\mu = \partial_\mu \delta_{ij} + i g_s \sum_A^{N_c^2-1} t_{ij}^A \mathcal{A}_\mu^A(x)$$ **new gluon field**

implies transformation: $\rightarrow t^A(\mathcal{A}_\mu^A)' = U(x) t^A \mathcal{A}_\mu^A U^{-1}(x) + \frac{i}{g_s}(\partial_\mu U(x)) U^{-1}(x)$

→ cancels unwanted terms, but since the gluon field has to be dynamic
need kinematic term for the gluon field itself!

Yang-Mills Theory

4. ingredient: dynamic term for the gluon fields
→ YM: Generalization of Maxwell equations to a general group

$$\mathcal{F}_{\mathrm{YM}} = -\frac{1}{4} F^A_{\mu\nu} F^{\mu\nu,A} \qquad F^A_{\mu\nu} = \partial_\mu \mathcal{A}^A_\nu - \partial_\nu \mathcal{A}^A_\mu - g_s f^{ABC} \mathcal{A}^B_\mu \mathcal{A}^C_\nu$$

Field strength tensor

This is absent in QED
→ gluons carry charge!

Structure constants: $[t^A, t^B] = i f^{ABC} t^C$

fully anti-symmetric & real
if non zero: non-abelian
→ transformations U do not commute

Gauge invariance can be easily seen through: $i g_s t^A F^A_{\mu\nu} = [D_\mu, D_\nu]$

To summarize…

Quantum Chromodynamics is a **local gauge theory** of six quarks $q = d, u, s, c, b, t$

$$\mathcal{L} = \bar{\Psi}_{q,a}(\underbrace{i\partial\!\!\!/\delta_{ab} - g_s t^A_{ab} A\!\!\!/^A}_{(D_\alpha)_{ab} = \partial_\alpha \delta_{ab} + i g_s (t^C A^C_\alpha)_{ab}} - m_q \delta_{ab})\Psi_{q,b} - \frac{1}{4} F^A_{\mu\nu} F^{A,\mu\nu} \qquad F^A_{\mu\nu} = \partial_\mu A^A_\nu - \partial_\nu A^A_\mu - g_s f^{ABC} A^B_\mu A^C_\nu$$

$SU(3)$ → 3 fundamental indices or ‘colours’ (index: a) (N_c)

$(SU(N_c))$ → 8 adjoint indices/generators or gluons (index: A) $\left(N_c^2 - 1\right)$

‘Colour’ algebra:

$$[t^A, t^B]_{ab} = i f^{ABC} t^C_{ab}$$

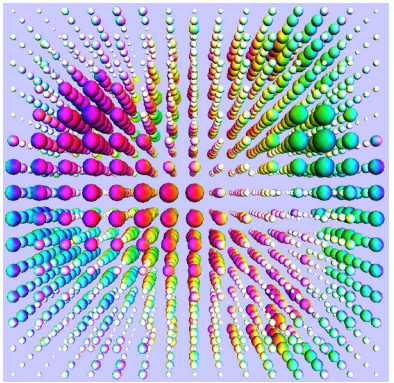
structure constants

$$\mathrm{Tr}\, t^A t^B = T_F \delta^{AB} \text{ with } T_F = \frac{1}{2}$$

normalization

How to solve QCD?

Two complementary approaches:



Lattice QCD

→ discretize space time and compute expectation values of operators

→ problem: the higher the energy the finer the lattice has to be
+ extrapolation / continuum limit

→ perfect for low energy: hadron spectrum, non-perturbative effects

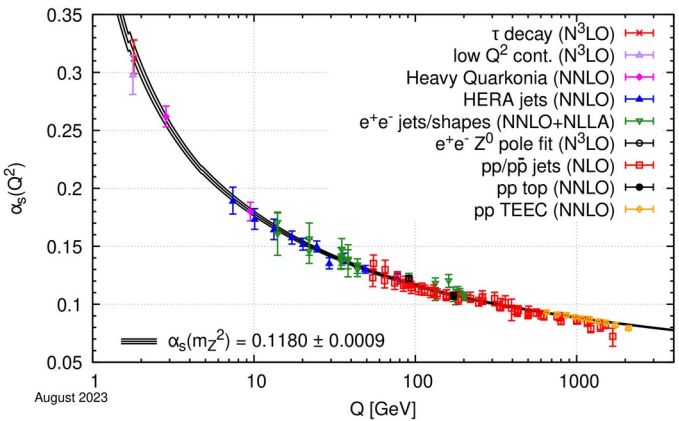
Perturbation theory

→ small coupling (implies high energy as we will see)

→ expansion in coupling constant → Feynman diagrams

$$\hat{\sigma}_{ab\to X} = \hat{\sigma}^{(0)}_{ab\to X} + \hat{\sigma}^{(1)}_{ab\to X} + \hat{\sigma}^{(2)}_{ab\to X} + \mathcal{O}(\alpha_s^3)$$

→ ideal for high energy scattering



The issue with the gluon propagator

Before we can write down the Feynman rules, we have to figure out a small problem with the gluon propagator…

Let's focus on: $\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4} F^a_{\mu\nu} F^{a,\mu\nu}$

Propagator: inverse of gaussian term of the action

$$S_{\mathrm{YM}} = i \int \mathrm{d}^4x \, -\frac{1}{4} F^a_{\mu\nu} F^{a,\mu\nu} \ni \frac{i}{2} \int \mathrm{d}^4x A^a_\mu (\partial^2 g^{\mu\nu} - \partial^\mu \partial^\nu) A^b_\nu \delta_{ab} \quad \sim \frac{1}{2} \int \frac{\mathrm{d}^4k}{(2\pi)^4} \tilde{A}^a_\mu(p) (p^2 g^{\mu\nu} - p^{\mu\nu}) \tilde{A}^b_\nu(p) \delta_{ab}$$

$$i\Delta_{\mu\rho} \left(p^2 g^{\rho\nu} - p^\rho p^\nu\right) = g^\nu_\mu \qquad \text{but} \qquad i\Delta_{\mu\rho} \left(p^2 g^{\rho\nu} - p^\rho p^\nu\right) p_\nu = 0 \neq p_\mu$$

Needs an additional constraint on the gluon field! → Gauge fixing

For example **covariant gauges**: $\partial \mathcal{A} = 0 \quad \rightarrow \mathcal{L}_{\mathrm{GF}} = -\frac{1}{2\lambda} \left(\partial_\mu \mathcal{A}^\mu\right)^2 \quad \lambda \in \mathbb{R}$

Lagrange multiplier methods for constraints

Gluon propagator in covariant gauge

$$i\Delta_{\mu\rho}\left(p^2 g^{\rho\nu} - (1-\frac{1}{\lambda})p^\rho p^\nu\right) = g_\mu^\nu \quad \rightarrow \quad \Delta_{\mu\nu} = \frac{-i}{p^2+i\epsilon}\left[g_{\mu\nu} - (1-\lambda)\frac{p^\mu p^\nu}{p^2}\right]$$

$\lambda = 0$ Landau gauge

$\lambda = 1$ Feynman gauge

But generally a good tool to check Independence on λ of final results!

A small issue remains: non-transvers [remember gluons are massless vectors] can propagate
→ using ghosts fields to cancel these! [L. Faddeev, V. Popov 1967]

Faddeev-Popov matrix in covariant gauge:

$$\mathcal{L}_{\mathrm{FP}} = \eta_A^\dagger M^{AB} \eta_B \qquad M^{AB} = \delta^{AB}\partial_\mu\partial^\nu + g_s f^{ABC}\mathcal{A}_\mu^C\partial^\mu$$

coloured complex scalar fields, do not appear as external states
→ needs to new interaction and needs to be included in amplitude computations!

Physical gauges

Axial gauge: $n_\mu A^\mu = 0 \qquad p \cdot n \neq 0$ abritary vector

($n^2 = 0$ called light-cone gauge)

$$\rightarrow \mathcal{L}_{\mathrm{GF}} = -\frac{1}{2\lambda}\left(n_\mu \mathcal{A}^\mu\right)^2$$

Gluon propagator: $\Delta_{\mu\nu}(p,n) = \frac{-i}{p^2 + i\epsilon}\left[g_{\mu\nu} - \frac{p_\mu n_\nu + p_\nu n_\mu}{p\cdot n} + \frac{n^2 p_\mu p_\nu}{(p\cdot n)^2}\right]$

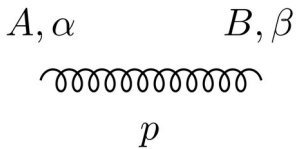
$$M^{AB} = \delta^{AB}\partial_\mu\partial^\nu + \underbrace{g_s f^{ABC} n^\mu \mathcal{A}^C_\mu}_{=0}$$

→ no propagating ghosts

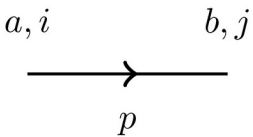
Implication in practise: more terms in gluon propagator but less diagrams

Perturbative QCD: Feynman rules (in covariant gauge)

Propagators

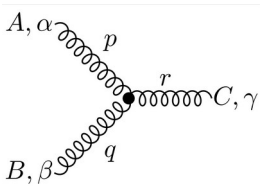


$$\delta^{AB}\left[-g^{\alpha\beta}+(1-\lambda)\frac{p^\alpha p^\beta}{p^2+i\epsilon}\right]\frac{i}{p^2+i\epsilon}$$

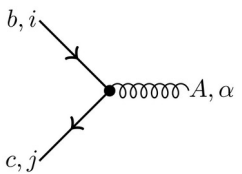


$$\delta^{ab}\frac{i}{(\not{p}-m+i\epsilon)_{ij}}$$

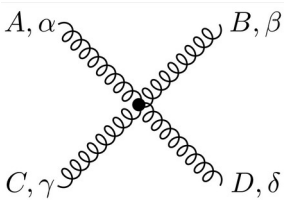
Vertices



$$-g_s f^{ABC}\left[(p-q)^\gamma g^{\alpha\beta}+(q-r)^\alpha g^{\beta\gamma}+(r-p)^\beta g^{\gamma\alpha}\right]$$



$$-ig_s t^A_{cb}\gamma^\alpha_{ji}$$



$$-ig_s^2 f^{XAC}f^{XBD}\left[g^{\alpha\beta}g^{\gamma\delta}-g^{\alpha\delta}g^{\beta\gamma}\right]+\dots$$

+ rule for loops: $\int \frac{\mathrm{d}^d l}{(2\pi)^d}$

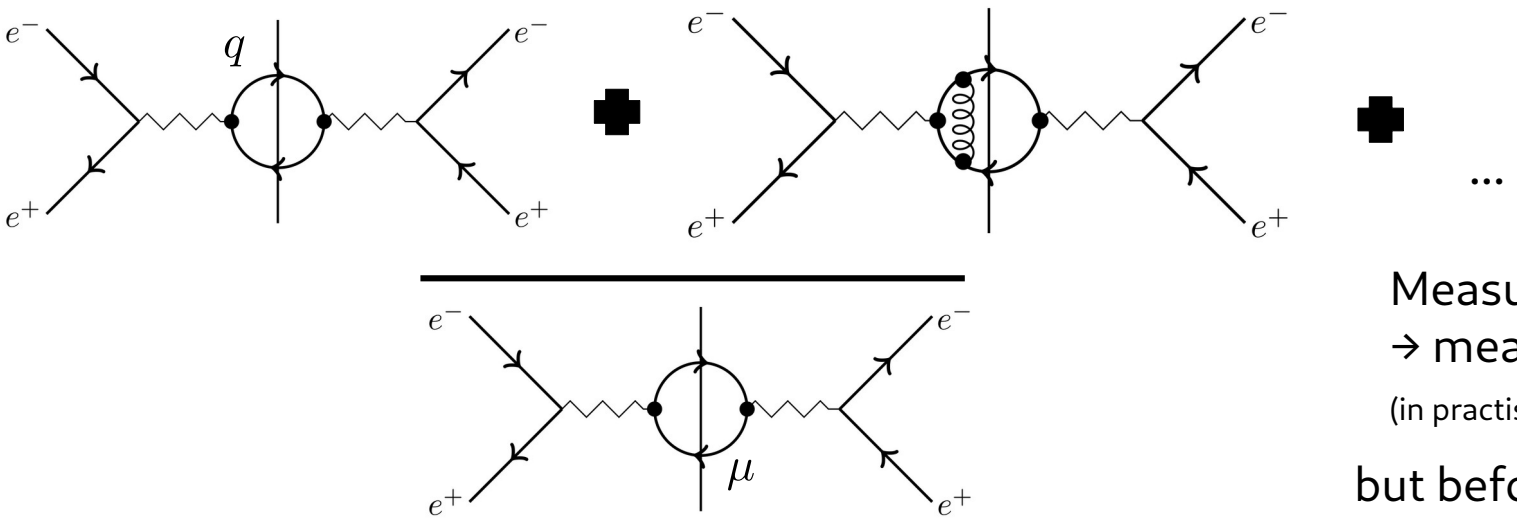
Ghost propagator: $g_s f^{ABC} q^\alpha$

Gluon-ghost vertex: $\delta^{AB}\frac{i}{p^2+i\epsilon}$

The strong coupling constant

In **massless** QCD the strong coupling constant is the **only free** parameter: $g_s = \sqrt{\alpha_s 4\pi}$
→ determine it through experiement, a possible idea would be

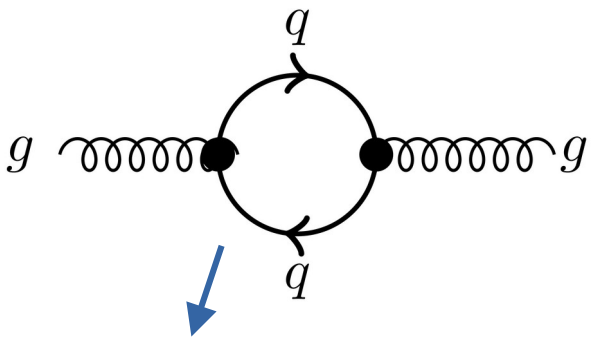
$$R = \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)} = \left[N_c \sum_q Q_q^2\right]\left(1 + \frac{\alpha_s}{\pi} + \dots\right)$$

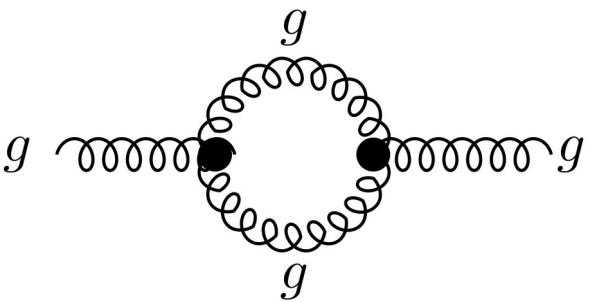


Measurement of R
→ measurement of α_s
(in practise not the best observable)

but before we do this….

UV renormalisation of the strong coupling





ghosts

$$\sim \int \frac{\mathrm{d}^4 q}{(2\pi)^4} \frac{\mathrm{Tr}\left[\gamma^\mu (q+m)\gamma^\nu (q+k+m)\right]}{(q^2-m^2)[(q+k)^2-m^2]} \quad \xrightarrow{|q|\to\infty} \quad \int \mathrm{d}q \frac{1}{q}$$

Ultra-violet divergence

Consider **cut-off regularization**: $\int_{q<\Lambda} \frac{\mathrm{d}q}{q} \sim \ln \Lambda$

Recover the full theory in the limit $\Lambda \to \infty$ breaks Lorentz invariance :(

Dimensional regularization

Working with a cut-off is cumbersome: broken Lorenz invariance of amplitudes...

Commonly used alternative: **dimensional regularization**

→ working in $d = 4 - 2\varepsilon$

→ Keeps **Lorentz** and **gauge invariance**

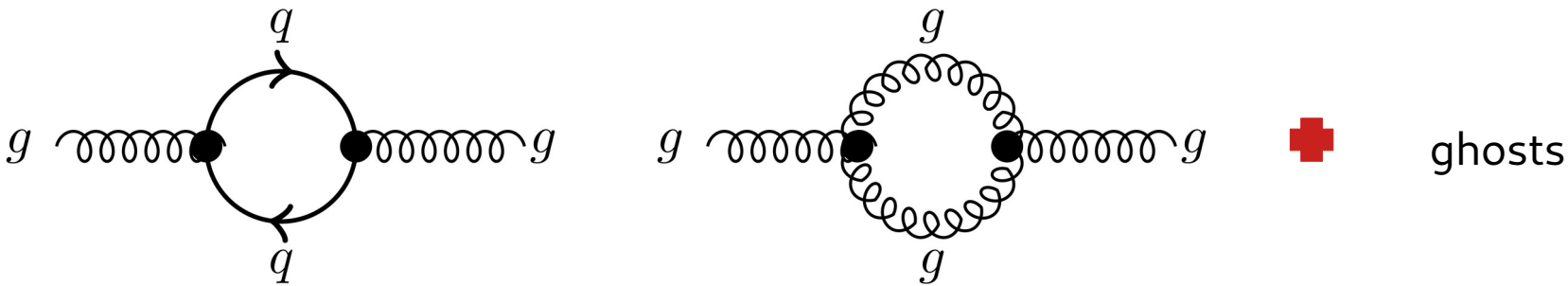
→ **Infrared divergences** can be treated in the same way

→ Instead of logarithms we find poles: $\frac{1}{\epsilon^n}$ for $\epsilon \to 0$

→ Implementation:

- → modify momentum integrals $\frac{\mathrm{d}^4 q}{(2\pi)^4} \to \frac{\mathrm{d}^d q}{(2\pi)^d}$
- → modified Lorentz and Dirac-algebra $\mathrm{Tr}\, g^{\mu\nu} = -2(1-\varepsilon)$
- → rescaled coupling (dimensionless) $g_s \to \mu^\varepsilon g_s$

Renormalisation of the strong coupling



ghosts

$$\Pi^{gg}(k^2) = \frac{\alpha_s^{\text{bare}}}{2\pi}\left[-\beta_0\left(\frac{1}{\varepsilon} - \gamma_E + \ln(4\pi) + \ln\left(\frac{\mu^2}{k^2}\right)\right) + \mathcal{O}(\varepsilon)\right] \qquad \beta_0 = \frac{11}{6}C_A - \frac{1}{3}n_f$$

Renormalization in QFTs:

- **bare Lagrangian parameters** are **not physical** quantities
- **absorb** (UV) divergences in **parameter definition** $g_s^{\text{bare}} = Z_g g_s$
- **physical measurements** fix the renormalized value

UV renormalization in QCD

$\overline{\mathrm{MS}}$ Scheme (massless QCD, covariant gauge):

Wave functions:

$$Z_q = 1 - \lambda C_F \frac{\alpha_s S_\varepsilon}{4\pi\varepsilon} + \mathcal{O}(\alpha_s^2)$$

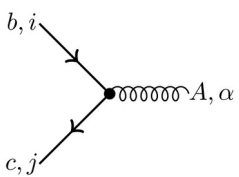
$$Z_g = 1 - \frac{\alpha_s S_\varepsilon}{4\pi\varepsilon}\left[\left(\frac{\lambda}{2} - \frac{13}{6}\right) C_A + \frac{4}{3} T_F n_f\right] + \mathcal{O}(\alpha_s^2)$$

Coupling:

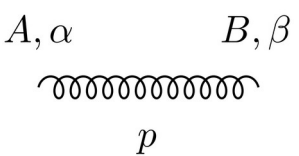
$$Z_{g_s} = 1 - \frac{\alpha_s S_\varepsilon}{4\pi\varepsilon} \frac{11 C_A - 4 n_f T_F}{6} + \dots$$

$$S_\varepsilon = \frac{(4\pi)^\varepsilon}{\Gamma(1-\varepsilon)}$$

These introduce new diagrams in the perturbative expansions:
→ cancellation of all UV divergences



$\times (Z_{g_s} - 1)$



$\times (Z_g - 1)$

Renormalization group equations

The Lagrangian parameter now depend on an arbitrary scale μ

→ **physical quantities do not, for example the R-ratio**

$$\mu^2 \frac{\mathrm{d}}{\mathrm{d}\mu^2} R\left(\frac{Q^2}{\mu^2}, \alpha_s\right) \equiv \left[\mu^2 \frac{\partial}{\partial \mu^2} + \mu^2 \frac{\partial \alpha_s}{\partial \mu^2} \frac{\partial}{\partial \alpha_s}\right] R = 0 \qquad \left[-\frac{\partial}{\partial t} + \beta(\alpha_s) \frac{\partial}{\partial \alpha_s}\right] R(e^t, \alpha_s) = 0$$

$$t = \int_{\alpha_s}^{\alpha_s(Q^2)} \frac{\mathrm{d}x}{\beta(x)}, \quad \alpha_s(\mu^2) = \alpha_s \longrightarrow R(1, \alpha_s(Q^2))$$

scale dependence only through α_s

The running coupling

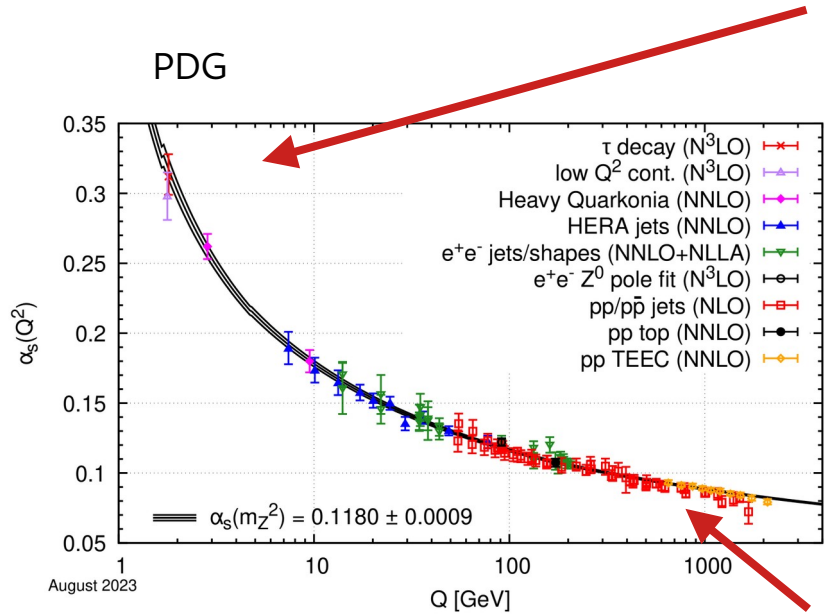
Beta-function: $\mu^2 \frac{\partial \alpha_s}{\partial \mu^2} = \beta(\alpha_s) \quad \beta(\alpha_s) = -\alpha_s^2(b_0 + b_1\alpha_s + \dots) \qquad b_0 = \frac{11C_A - 2n_f}{12\pi}$

The renormalised coupling is fixed by experiment where we identify a reference scale

For example at the Z-pole: $\alpha_s^{\text{measured}} \equiv \alpha_s(m_Z^2)$

First order solution: $\alpha_s(\mu^2) = \frac{\alpha_s(m_Z^2)}{1 + \beta_0 \frac{\alpha_s(m_Z^2)}{2\pi} \ln \frac{\mu^2}{m_Z^2}}$

Asymptotic freedom & confinement

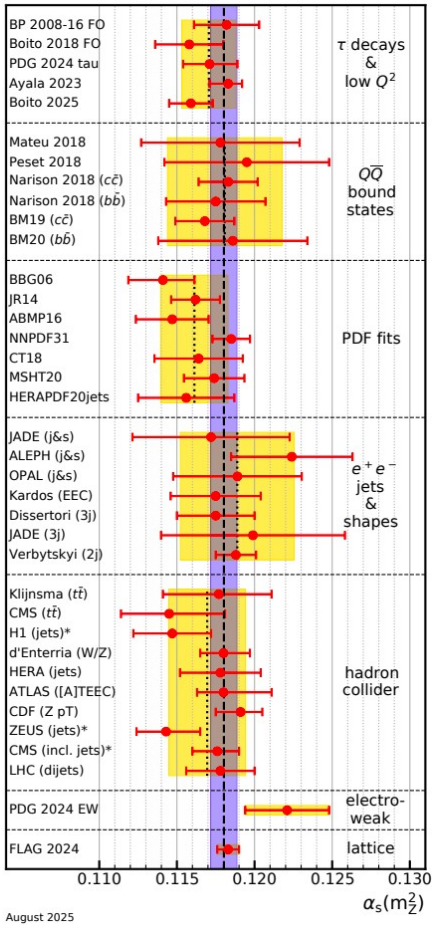


Confinement at small energy scales:
→ perturbation theory breaks downs
→ QCD bound states

$$\alpha_s(\mu^2) = \frac{\alpha_s(m_Z^2)}{1 + \beta_0 \frac{\alpha_s(m_Z^2)}{2\pi} \ln \frac{\mu^2}{m_Z^2}}$$

Asymptotic freedom at high energies:
→ dynamics of individual quarks and gluons
→ good regime for perturbation theory

Strong coupling determinations



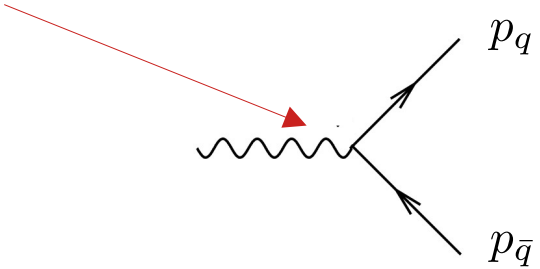
- Various ways to extract the strong coupling from data
  - PDF fits
  - τ-decays
  - e+e- hadronic final states
  - Hadron collider → jets or top-quarks
  - Electroweak precision fit
  - Lattice QCD
- Overall consistent picture!
- But there some challenges:
  - Higher-order QCD corrections are crucial
  - Non-perturbative effects dilute the picture even at high energies
    → control over systematics (TH & EXP) essential

Final state radiation – Infrared divergencies

Gluon emission from a quark-anti-quark pair

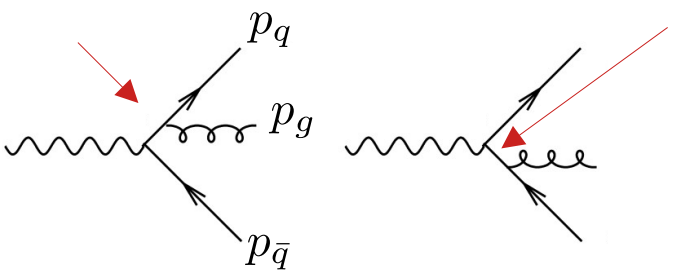
$$\mathcal{M}^{\mu}_{q\bar{q}} = \bar{u}(p_q)\,(-ie\delta_{ab}\gamma^{\mu})\,v(p_{\bar{q}})$$

[Note: suppressed open colour and spin indices]



Let's consider a gluon emission:

$$\mathcal{M}^{\mu}_{q\bar{q}g} = \bar{u}(p_q)(-ig_s t^A_{ab}\not{\epsilon})\frac{i(\not{p}_q + \not{p}_g)}{(p_q+p_g)^2}(-ie\gamma^{\mu})v(p_{\bar{q}}) + \bar{u}(p_q)(-ie\gamma^{\mu})\frac{i(\not{p}_{\bar{q}} - \not{p}_g)}{(p_{\bar{q}}-p_g)^2}(-ig_s t^A_{ab}\not{\epsilon})v(p_{\bar{q}})$$



The soft limit

Extract leading behaviour in the soft limit:

$$\ldots \bar{u}(p_q)(-ig_s t^A_{ab} \not{\epsilon}) \frac{i(\not{p}_q + \not{p}_g)}{(p_q + p_g)^2} (-ie\gamma^\mu) v(p_{\bar{q}}) + \ldots$$

$\mathcal{O}(\lambda^{-1})$ $\mathcal{O}(\lambda^0)$

Parameterize limit: $p_g \to \lambda p_g$, $\lambda \to 0$

A bit of algebra later:

$$\mathcal{M}^\mu_{q\bar{q}g} \to \underbrace{\bar{u}(p_q)(-ie\gamma^\mu)v(p_{\bar{q}})}_{\mathcal{M}^\mu_{q\bar{q}}} (-ig_s t^A) \left(\frac{p_q \cdot \epsilon}{p_q \cdot p_g} - \frac{p_{\bar{q}} \cdot \epsilon}{p_{\bar{q}} \cdot p_g} \right)$$

Some handy identities:

$$(p_q + p_g)^2 = 2p_q \cdot p_g$$

$$\not{p}_{\bar{p}} v(p_{\bar{p}}) = 0$$

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}$$

Soft/Eikonal factor
→ this factor is a general feature of QCD
→ Soft/collinear factorization!

The soft limit of the cross section

Cross sections are defined in terms of the squared matrix element and the phase space
→ sum over unobserved degrees of freedom: spin and color indicies

$$\mathrm{d}\Phi_3 \sum_{\mathrm{pol}} |\mathcal{M}_{q\bar{q}g}| = \mathrm{d}\Phi_{q\bar{q}} |M_{q\bar{q}}|^2 \mathrm{d}\Phi_g C_F g_s^2 \frac{2p_q \cdot p_{\bar{q}}}{(p_q \cdot p_g)(p_{\bar{q}} \cdot p_g)}$$

$$\mathrm{d}\Phi_3 = \frac{\mathrm{d}\vec{p}_q}{2E_q(2\pi)^3} \frac{\mathrm{d}\vec{p}_{\bar{q}}}{2E_{\bar{q}}(2\pi)^3} \frac{\mathrm{d}\vec{p}_g}{2E_g(2\pi)^3} \delta^4(Q - p_q - p_{\bar{q}} - p_g)$$

$$\to \frac{\mathrm{d}\vec{p}_q}{2E_q(2\pi)^3} \frac{\mathrm{d}\vec{p}_{\bar{q}}}{2E_{\bar{q}}(2\pi)^3} \delta^4(Q - p_q - p_{\bar{q}}) \left[\frac{\mathrm{d}\vec{p}_g}{2E_g(2\pi)^3} \right]$$

$$\mathrm{Tr}\, t^A t^A = T_F \delta^{AA} = N_C C_F$$

$$\sum_{\mathrm{pol}} \epsilon^\mu \epsilon^\nu = -g^{\mu\nu} \dots$$

Parameterize in terms of angles and energies:

$$p_q \cdot p_g = E_q E_g (1 - \cos\theta)$$
$$p_{\bar{q}} \cdot p_g = E_{\bar{q}} E_g (1 + \cos\theta)$$

$$\frac{|\mathcal{M}_{q\bar{q}g}|^2 \mathrm{d}\Phi_3}{|\mathcal{M}_{q\bar{q}}|^2 \mathrm{d}\Phi_2} = \frac{2\alpha_s C_F}{\pi} \frac{\mathrm{d}E_g}{E_g} \frac{\mathrm{d}\theta}{\theta} \frac{\mathrm{d}\phi}{2\pi}$$

Soft and collinear divergencies

$$\frac{|\mathcal{M}_{q\bar{q}g}|^2 \mathrm{d}\Phi_3}{|\mathcal{M}_{q\bar{q}}|^2 \mathrm{d}\Phi_2} = \frac{2\alpha_s C_F}{\pi} \frac{\mathrm{d}E_g}{E_g} \frac{\mathrm{d}\theta}{\sin(\theta)} \frac{\mathrm{d}\phi}{2\pi}$$

$E_g \to 0$ Soft divergence

$\theta \to 0, \quad \theta \to \pi$ Collinear divergencies wrt to quark or anti-quark

Ultimately these will cancel against virtual corrections → see tomorrow

Integrate over phase space (with a suitable cutoff $E_g\theta \sim k_T$):

$$\frac{2\alpha_s C_F}{\pi} \int \frac{\mathrm{d}E_g}{E_g} \int \frac{\mathrm{d}\theta}{\sin(\theta)} \theta(E_g\theta - \lambda) \sim \frac{2\alpha_s C_F}{\pi} \log^2 \lambda$$

'double-logarithm'

Note: similar for gluon emissions from gluons but with $C_F \to C_A$

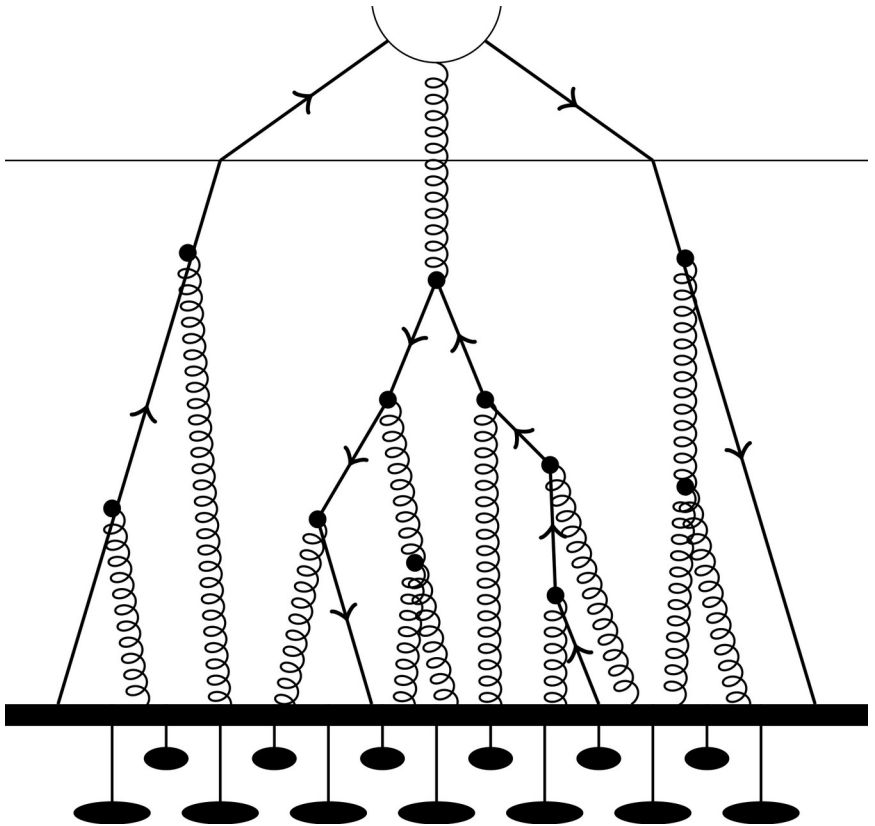
Iterated emissions

→ the probability to emit a gluon is enhanced in the soft and collinear regions

→ expect a cascade of emissions

→ our treatment fails at very low scales i.e. perturbation theory breaks down

→ transition to non-perturbative regime: formation of hadrons



End of lecture 1 - summary

- A bit of history…
- Structure of QCD as a SU(3) local gauge theory
- UV Renormalization
- Running of the strong coupling constant
- Soft/collinear divergencies for gluon emissions